Zizhong Wang

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Education

North Carolina State University Raleigh, NC
M.S. in Computer Science; GPA: 3.83/4.00 May 2027 (expected)
Relevant Coursework: GPU Architecture (ECE 786); Parallel Computer Architecture and Programming (CSC 506); Optimization and Algorithms (CSC 572).

East China Normal University Shanghai, China
B.Eng. in Software Engineering Sep 2021 – Jun 2025
Honors: Third Class Scholarship, 2024

Research Interests

AI Infrastructure; Algorithm–System Co-design for Large-Scale Model Training and Inference; GPU and Distributed Systems.

Publications

PuzzleKV: Page-Wise Low-Rank Decomposition for KV Cache Compression
Zizhong Wang, Jieying Wang, Zhao Zhang, and Jiajia Li.
arXiv preprint, 2026. [arXiv] · *Under review.*

GeoFP8: Geometry-Aware FP8 Gradient Compression for Distributed LLM Training
Jieying Wang, **Zizhong Wang**, Fangru Linghu, Shuyuan Fan, Jiajia Li, and Zhao Zhang.
arXiv preprint, 2026. [arXiv] · *Under review.*

Research Experience

North Carolina State University Raleigh, NC
Graduate Researcher, advised by Prof. Jiajia Li Aug 2025 – Present
Graduate Research Assistant, Summer 2026

PuzzleKV: Page-Wise Low-Rank KV Cache Compression
Apr 2026 – Aug 2026

- Developed the core algorithm and system design for page-wise low-rank KV cache compression, aligning compression granularity with paged KV cache management.
- Designed direct attention over compressed factors without KV reconstruction, including fused attention kernels for mixed dense and factorized KV pages.
- Implemented **batched Gram-matrix eigendecomposition** for page-wise SVD and incremental page compression during decoding, avoiding recomputation of previously compressed pages.
- Preserved **98.7%** of Full-KV performance on LongBench and **96.2%** on RULER at **~60% KV storage**; with INT4 factor quantization, retained **93.4%** on RULER at **18.7% storage**.

GeoFP8: Geometry-Aware FP8 Gradient Compression for Distributed LLM Training
May 2026 – Sep 2026
Collaborated with Prof. Zhao Zhang, Rutgers University

- GeoFP8 transforms anisotropic gradients using approximate Fisher geometry to reduce quantization distortion during low-precision communication.
- Contributed to GeoFP8's integration into Megatron-LM and implemented a staged asynchronous pipeline overlapping FP8 AllReduce with gradient preparation and dequantization/mapback.
- Built controlled multi-GPU scaling experiments; on **64 NVIDIA GH200 GPUs**, GeoFP8 reduced end-to-end pretraining time for Llama-600M by **7.6%** relative to FP32 gradient communication.

***KronWeave*: CP-Structured Kronecker Generation for Power-Law Sparse Tensors**
Jan 2026 – Apr 2026

- Developed a parameterized CP-structured Kronecker generator for controllable power-law sparse tensors, motivated by structural artifacts in manually designed Kronecker initiators.
- Evaluated generated tensors using mode-wise degree, cross-mode, and multiscale structural diagnostics.
- Contributed KronWeave to ***TensorSuite***, an open suite of sparse tensor datasets and generation tools, and co-developed parts of the project website.

Low-Precision Optimizer and Gradient Communication for LLM Pretraining
Aug 2026 – Present

- Investigating NVFP4-based optimizer updates and gradient communication for efficient distributed LLM pretraining.
- Developing and profiling GPU- and communication-aware implementations to evaluate numerical stability, communication efficiency, and end-to-end training performance.

Technical Blog

Softmax Collapse Beyond Adam: What Happens to Muon? [Blog]
An experimental study of softmax collapse in Muon, examining gradient geometry, momentum spectra, and the effects of numerically stable cross-entropy on training dynamics.

Professional Experience

Shanghai Shuhe Technology Shanghai, China
Software Engineering Intern Jul – Sep 2024

- Contributed to production systems supporting customer-service and membership operations, helping reduce customer-service costs and support requests by approximately 50%.
- Designed configurable internal tools and improved engineering workflows, identifying and proposing 15+ solutions to usability and performance bottlenecks.
- Authored internal technical documentation and contributed to shared development practices, supporting effective collaboration and knowledge transfer.

Technical Skills

Programming: Python, C/C++
GPU Programming: CUDA, CuTe, Triton
ML Systems: PyTorch, Megatron-LM, vLLM
Distributed & HPC: NCCL, Slurm, Linux
Profiling & Development: Nsight Systems, Nsight Compute, Git